

Probabilities. Classical approach

Counting techniques

► arrangements of n distinct objects taken k at a time

- n people are seated around a table with exactly $k \leq n$ seats; *the order matters*, thus with k people already arranged one can form another $k!$ distinct arrangements
- the number of all possible arrangements

$$A_n^k = \frac{n!}{(n-k)!}$$

► arrangements with repetition

- we create words of length k with distinct letters from an alphabet with $n \geq k$ letters; the order matters and we can *use the same letter multiple times*
- we can create exactly

$$A_{rep,n}^k = n^k \quad \text{words}$$

► combination of n distinct objects taken k at a time

- we have a set consisting of n elements and we want to form a subset consisting of k elements; obviously in a subset the *order doesn't matter*, only the chosen elements are important
- the number of possibilities is

$$C_n^k = \frac{n!}{(n-k)! \cdot k!} \quad \text{subsets}$$

► combinations with repetitions

- we place n identical objects into k boxes; *some of these boxes can stand empty*, the order of these boxes is predetermined
- equivalently, we search for all natural solutions of the equation

$$x_1 + x_2 + \dots + x_k = n$$

- the answer is the same

$$C_{rep,n}^k = \frac{(n+k-1)!}{k!(n-1)!} \quad \text{possibilities}$$

► the bijection principle

- two finite sets A and B have the *same number of elements* if and only if there exists a bijective function $f: A \rightarrow B$.
- it will be very useful in applications to find an equivalent bijective translation of a given problem, most of the times using the patterns defined here

► **arrangements of n different objects into k boxes:** there are given n different objects and k different boxes $B_1, B_2, \dots, B_k$ and we have to place n_1 objects into the box B_1 , n_2 objects into the box B_2 , etc., and n_k objects into the box B_k , with $n_1 + n_2 + \dots + n_k = n$

- suppose that the order of elements in the individual boxes doesn't matter but the **order of these boxes is important** in the following way: we not just place n_1 objects into a box, we place n_1 into the first box, hence we have a predefined order of these boxes which is extremely important in applications.
- the number of all such distributions is the multinomial coefficient

$$\frac{n!}{n_1! n_2! \cdot \dots \cdot n_k!} \quad n_1 + n_2 + \dots + n_k = n.$$

► **the inclusion-exclusion principle:** for any finite sequence $A_1, A_2, \dots, A_n$ of subsets of a finite set X the number of elements in their union is

$$\left| \bigcup_{k=1}^n A_k \right| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \dots + (-1)^{p-1} \cdot \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_p}| + \dots + (-1)^{n-1} \cdot |A_1 \cap A_2 \cap \dots \cap A_n|$$

Classical definition of probability

The probability of an event is the ratio of the number of cases favorable to it, to the number of all cases possible when nothing leads us to expect that any one of these cases should occur more than any other, which renders them, for us, equally possible. (Pierre Simon Laplace)

► **Poincare's theorem:** the probability of the union of n events is

$$P\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n P(E_k) - \sum_{k < j} P(E_k \cap E_j) + \sum_{k < j < i} P(E_k \cap E_j \cap E_i) - \dots + (-1)^{n-1} P(E_1 \cap E_2 \dots \cap E_n)$$

► the probability of the intersection is given by **the multiplication rule**

$$P\left(\bigcap_{k=1}^n E_k\right) = P(E_1) \cdot P(E_2|E_1) \cdot P(E_3|E_1 \cap E_2) \cdot \dots \cdot P\left(E_n \left| \bigcap_{k=1}^{n-1} E_k \right.\right)$$

if the events are independent the above formula becomes

$$P\left(\bigcap_{k=1}^n E_k\right) = P(E_1) \cdot P(E_2) \cdot \dots \cdot P(E_n).$$

► **more about conditional probabilities:** the probability that an event E will occur simultaneously with one of the events $H_1, H_2, \dots, H_n$, that form a system of *mutually exclusive and exhaustive* events (hypotheses), is given by the **total probability formula**

$$P(E) = \sum_{k=1}^n P(H_k) \cdot P(E|H_k)$$

where $\sum_{k=1}^n P(H_k) = 1$

- the probability $P(H_j|E)$ of the hypothesis H_j if the event E occurred is given by the **Bayes' formula**

$$P(H_j|E) = \frac{P(H_j) \cdot P(E|H_j)}{\sum_{k=1}^n P(H_k) \cdot P(E|H_k)}$$

Some probability schemes

► **the binomial experiment:** is a statistical experiment that has the following properties

- the experiment consists of n repeated *trials*
- each trial can result in just two possible outcomes: a *success* and a *failure*
- the probability of success, denoted by p , and the probability of failure, denoted $q = 1 - p$, are the same on every trial
- the trials are independent
- the **binomial probability** refers to the probability that a binomial experiment results in exactly k successes out of n trials

$$P = C_n^k \cdot p^k \cdot q^{n-k}$$

- the probability that the k -th success is obtained after exactly r trials is

$$P = C_{r-1}^{k-1} \cdot p^k \cdot (1-p)^{r-k}, \quad r \geq k.$$

► **the multinomial experiment:** generalizes the binomial experiment, now each trial can have k outcomes: $E_1, E_2, \dots, E_k$

- the corresponding probabilities are $p_1, p_2, \dots, p_k$
- the n trials are again independent.
- the **multinomial probability** is the probability that E_1 occurs n_1 times, E_2 occurs n_2 times, $\dots$ E_k occurs n_k times

$$P = \frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!} \cdot p_1^{n_1} \cdot p_2^{n_2} \cdot \dots \cdot p_k^{n_k}$$

where $n = n_1 + n_2 + \dots + n_k$

► **Poisson's scheme**

- let $E_1, E_2, \dots, E_n$, be n independent events of an experiment.
- denote by p_i the probability of E_i to occur and by $q_i = 1 - p_i$, $i = \overline{1, n}$ the probability of the complementary event.
- the probability that k events, out of those n , occur is given by the coefficient of X^k in the expression

$$(p_1X + q_1) \cdot (p_2X + q_2) \cdot \dots \cdot (p_nX + q_n)$$